Conservation of Energy

Exam code: 0625 & 0972

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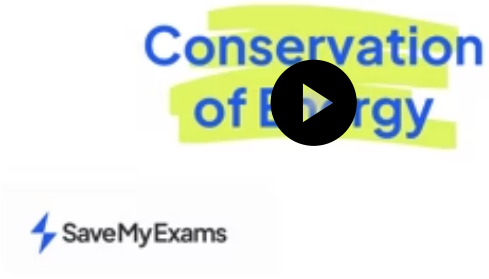

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Conservation of energy

- The **principle of conservation of energy** states that:

Energy cannot be created or destroyed, it can only be transferred from one store to another

- The principle of conservation of energy means that for a closed system, the total amount of energy is **constant**
- The **total** amount of energy transferred **into** the system must be **equal** to the **total** amount of energy transferred **away** from the system
- Therefore, energy cannot be 'lost', but it can be transferred to the surroundings
  - Energy can be **dissipated** (spread out) to the surroundings by heating and radiation
  - Dissipated energy transfers are often not **useful**, in which case they can be described as **wasted** energy

Examples of the principle of conservation of energy

Example 1: a bat hitting a ball

- The moving bat has energy in its **kinetic** store
- Some of that energy is transferred **usefully** to the **kinetic** store of the ball
- Some of that energy is transferred from the **kinetic** store of the bat to the **thermal** store of the ball **mechanically** due to the impact of the bat on the ball

Help me understand

Stuck on something?

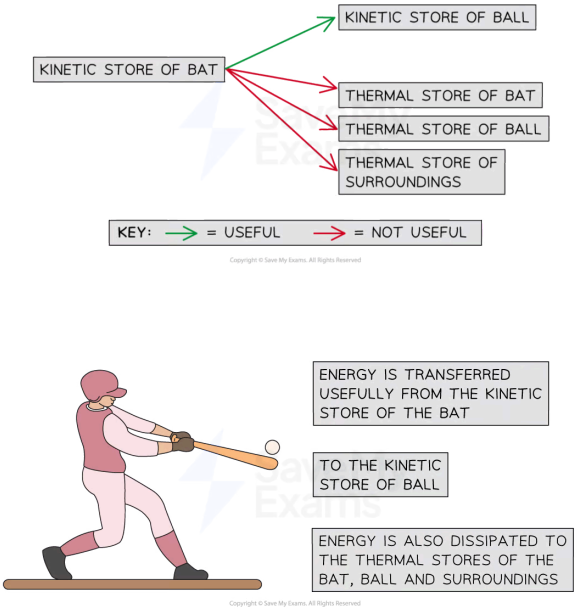
Ask questions and get help with exactly what you're revising.

Got it

- This energy transfer is not useful; the energy is **wasted**

- The total amount of energy transferred into the system is equal to the total amount of energy transferred away from the system

Conservation of energy: a bat hitting a ball



The principle of conservation of energy applied to a bat hitting a ball

Example 2: Boiling Water in a Kettle

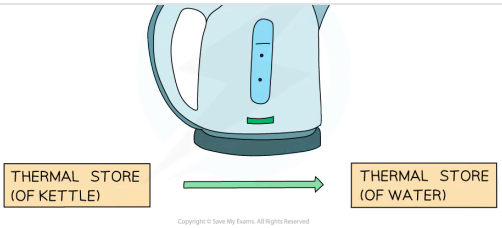
- When an electric kettle boils water, **energy** is transferred **electrically** from the mains supply to the **thermal store** of the heating element inside the kettle
- As the heating element gets hotter, **energy** is transferred **by heating** to the **thermal store** of the water

- Some of the energy is transferred to the **thermal** store of the plastic kettle
  - This energy transfer is not useful; the energy is **wasted**

- And some energy is **dissipated** to the **thermal store** of the surroundings due to the air around the kettle being heated
  - This energy transfer is not useful; the energy is wasted

- The total amount of energy transferred into the system is equal to the total amount of energy transferred away from the system

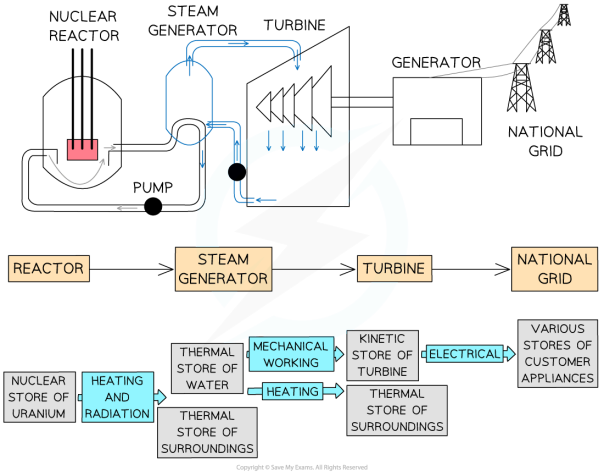
Conservation of energy: a kettle boiling water



The principle of conservation of energy applied to a kettle boiling water

Energy flow diagrams

- Energy stores and transfers can be represented using a flow diagram
  - This shows both the stores and the transfers taking place within a system



Energy flow diagram showing energy stores and transfers in a nuclear power plant.

Note the colour difference of the labels (stores) and the arrows (transfer pathways)

- In an energy flow diagram, energy is always **conserved**

total energy in = total energy out

Worked Example

The diagram shows a rollercoaster going down a track.

The rollercoaster takes the path A → B → C → D.

Which statement is true about the energy changes that occur for the rollercoaster down this track?

A. $E_K \rightarrow \Delta E_P \rightarrow \Delta E_P \rightarrow E_K$

B. $E_K \rightarrow \Delta E_P \rightarrow E_K \rightarrow \Delta E_P$

C. $\Delta E_P \rightarrow E_K \rightarrow E_K \rightarrow \Delta E_P$

D. $\Delta E_P \rightarrow E_K \rightarrow \Delta E_P \rightarrow E_K$

Answer: D

- **At point A:**
  - The rollercoaster is raised above the ground, therefore it has energy in its **gravitational potential store**
  - As it travels down the track, energy is transferred **mechanically** to its **kinetic store**

- **At point B:**
  - Energy is transferred **mechanically** from the kinetic store to the **gravitational potential store**
  - As the kinetic energy store **empties**, the gravitational potential energy store **fills**

- **At point C:**
  - Energy is transferred **mechanically** from the gravitational potential store to the **kinetic store**

- **At point D:**
  - The flat terrain means there is no change in the amount of energy in its gravitational potential store, the rollercoaster only has energy in its **kinetic store**
  - The kinetic energy store is full

  - In reality, some energy will also be transferred to the thermal energy store of the tracks due to **friction**, and to the thermal energy store of the surroundings due to **sound**
  - We say this energy is **dissipated to the surroundings**
    - The total amount of energy in the system will be constant
    - Total energy in = total energy out

Examiner Tips and Tricks

transferred between stores.

You may not always be given the energy transfers happening in the system in your IGCSE exam questions.

By familiarising yourself with the energy stores and transfer pathways, you should be able to relate these to the situation presented in the question. For example, a ball rolling down a hill transfers energy from the ball's gravitational potential energy store to its kinetic energy store mechanically, while a spring transfers energy from its elastic potential energy store to its kinetic energy store mechanically.

Sankey diagrams

Extended tier only

- **Sankey diagrams** can be used to represent energy transfers
  - Sankey diagrams are characterised by arrows that split to show the proportions of the energy transfers taking place
- The different parts of the arrow in a Sankey diagram represent the different energy transfers:
  - The left-hand side of the arrow (the flat end) represents the energy transferred **into** the system
  - The straight arrow pointing to the right represents the energy that ends up in the desired store; this is the **useful energy output**
  - The arrows that bend away represent the **wasted energy**

Features of a Sankey diagram

Total energy in, wasted energy and useful energy out shown on a Sankey diagram

- The width of each arrow is **proportional** to the amount of energy being transferred
- As a result of the conversation of energy:

total energy in = total energy out

total energy in = useful energy out + wasted energy

- A Sankey diagram for a modern efficient light bulb will look very different from that for an old filament light bulb

Sankey diagrams for an energy efficient bulb and a filament bulb

Filament bulbs have a much greater proportion of wasted energy than modern energy efficient bulbs

Worked Example

An electric motor is used to lift a weight. The diagram represents the energy transfers in the system.

Calculate the amount of wasted energy.

Answer:

Step 1: State the conservation of energy

- Energy cannot be created or destroyed, it can only be transferred from one store to another

$$\text{total energy in} = \text{useful energy out} + \text{wasted energy}$$

Step 2: Rearrange the equation for the wasted energy

$$\text{wasted energy} = \text{total energy in} - \text{useful energy out}$$

Step 3: Substitute the values from the diagram

$$\text{wasted energy} = 500 - 120$$

$$\text{wasted energy} = 380\ \text{J}$$

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